

ADVANCED ANALYTICS, MACHINE LEARNING, DEEP LEARNING & STATISTICS MASTER INTERVIEW
QUESTION BANK (120 QUESTIONS)

1. Define Artificial Intelligence.
2. Differentiate AI, ML, and Deep Learning.
3. Explain Narrow AI, General AI, and Super AI.
4. What is supervised learning?
5. What is unsupervised learning?
6. What is reinforcement learning?
7. What is semi-supervised learning?
8. What is self-supervised learning?
9. What is bias in ML?
10. What is variance?
11. Explain bias-variance tradeoff.
12. What is overfitting?
13. What is underfitting?
14. What is cross-validation?
15. What is K-fold cross-validation?
16. What is stratified sampling?
17. What is bootstrapping?
18. What is data leakage?
19. What is feature engineering?
20. What is feature selection?
21. Filter vs wrapper methods.
22. What is dimensionality reduction?
23. What is normalization?
24. What is standardization?
25. What is one-hot encoding?
26. What is label encoding?
27. What is multicollinearity?
28. What is regularization?
29. Ridge vs Lasso.
30. What is Elastic Net?
31. Assumptions of Linear Regression.
32. Derive cost function of Linear Regression.
33. What is gradient descent?
34. Batch vs Stochastic vs Mini-batch GD.
35. What is learning rate?
36. What is convex function?
37. Why logistic regression used for classification?
38. What is sigmoid function?
39. What is log loss?
40. What is decision boundary?
41. What is confusion matrix?
42. Accuracy vs Precision vs Recall.
43. What is F1-score?
44. What is ROC curve?
45. What is AUC?
46. RMSE vs MAE.
47. What is R-squared?

48. Adjusted R-squared.
49. What is entropy?
50. What is information gain?
51. What is Gini index?
52. Why decision trees overfit?
53. What is pruning?
54. What is Random Forest?
55. What is bagging?
56. What is boosting?
57. What is Gradient Boosting?
58. What is XGBoost?
59. What is LightGBM?
60. What is CatBoost?
61. What is stacking?
62. What is KNN?
63. How to choose K in KNN?
64. What is Naive Bayes?
65. Explain Bayes theorem.
66. What is SVM?
67. What is kernel trick?
68. What is hyperparameter tuning?
69. GridSearch vs RandomSearch.
70. What is Bayesian optimization?
71. What is PCA?
72. Why eigenvalues important in PCA?
73. What is covariance matrix?
74. What is SVD?
75. What is K-Means?
76. Steps of K-Means.
77. What is Elbow method?
78. What is Silhouette score?
79. What is hierarchical clustering?
80. What is DBSCAN?
81. What is anomaly detection?
82. What is time series forecasting?
83. What is stationarity?
84. What is ARIMA?
85. What is autocorrelation?
86. What is perceptron?
87. What is activation function?
88. ReLU vs Sigmoid vs Tanh.
89. What is softmax?
90. What is backpropagation?
91. What is vanishing gradient?
92. What is exploding gradient?
93. What is batch normalization?
94. What is dropout?
95. What is CNN?
96. What is convolution operation?
97. What is padding?
98. What is stride?

99. What is pooling?
100. What is transfer learning?
101. What is RNN?
102. Why RNN fails?
103. What is LSTM?
104. What is GRU?
105. What is attention mechanism?
106. What is transformer architecture?
107. What is BERT?
108. What is GPT?
109. What is embedding?
110. What is fine-tuning?
111. What is prompt engineering?
112. What is RAG?
113. What is model drift?
114. What is data drift?
115. What is MLOps?
116. What is model registry?
117. How to deploy ML model?
118. How to handle imbalanced dataset?
119. Explain bias-variance mathematically.
120. Why cross-entropy preferred over MSE for classification?